

Research Assignment 1

Section A: Data Fundamentals [Answers]

① Database is an organised electronic collection of information so that it can be easily accessed, updated, and managed. Databases are important because modern organisations create large amounts of data and need to retrieve large records quickly and securely.

Example: Hospitals use databases to manage patients records, appointments and medical histories.

② Data warehouse is a central system that collects data from different sources and prepares it for reporting and analysis. Unlike operational databases that are mainly used for daily activities such as sales and transactions, data warehouse stores historical data to identify trends and patterns.

Example: Companies combine sales, finance,

and customer data in a warehouse (via ETL) so managers can run reports without disrupting operations

③ A data lake stores data in its original form - tables, JSON, images, videos. A warehouse only keeps cleaned, organised data. Businesses prefer data lake when they want to do AI or machine learning later and need the raw data available.

④ Data Mart is a smaller version of data warehouse that focuses on one department. Finance teams often use one that only has budgets, expenses and revenue so they can get numbers they need quickly.

⑤ Structured: neat tables with rows and columns (customer database)
Semi-structured: some organisation but not strict tables (JSON or XML files)

Unstructured: no fixed format (photos, videos, email)

⑥ Data modeling is planning how data will be organized and linked in a system. It stops duplication and makes the data more accurate and secure.

Two common types are relational (tables joined by keys) and dimensions (fact and dimension tables used in warehouses)

⑦ Data mining means digging through large amounts of data to find patterns or unusual things. Businesses use it to understand customers, spot fraud or improve services. Example, supermarkets analyse buying habits to recommend products. Banks use it to catch suspicious transactions.

⑧ ETL → Extract, Transform, load.

Extract: pull data from different places

Transform: clean data and make it consistent

Load: put data into a warehouse or other systems

ETL is important ^{because} data from different sources is often messy, and clean data gives better reports.

⑧ Database Management System is the software that lets you create, store, update, and protect data in a database. It controls who can access what.

⑩ CSV: simple tables, easy to share. It is used for quick sharing.

JSON: good for apps and websites talking to each other. Used for APIs.

Avro: used when moving structured data between systems. It is used in data pipelines.

⑪ AI is technology that lets machines perform tasks that normally need human intelligence.

Narrow AI (Weak AI) is designed for one specific job (e.g. facial recognition).

General AI (Strong AI) would be able to do any intellectual task a human can - it does not exist yet.

⑫ Machine learning is when computer learn patterns from data instead of following fixed rules.

Traditional programming gives exact instructions. ML finds the patterns itself.

Supervised: learns from labelled examples

Unsupervised: finds patterns in ^{unlabelled data} ~~unlabeled data~~ ~~examples~~

Reinforcement: learns by trial and error. with reward

⑬ Deep learning is a type of machine learning that uses neural networks with many layers.

It sits inside ML, which sits inside AI.

Neural networks are inspired by the brain, - they have layers of connected nodes that process information and learn from examples.

⑭ NLP is technology that lets computers understand and work with human language. Everyday examples: voice assistants (Siri/Google), spam filters in emails, and auto-complete or translation on phones.

⑮ A large language model is an AI trained on huge amounts of text so it can generate and understand language. Standard ML models are usually smaller and built for one specific task. Well-known examples: GPT models (ChatGPT) for conversation and writing, and Claude for helpful text generation.

⑯ Generative AI creates new content (text, images, code, etc) based on what it has learned. At a high level it predicts the next piece of content from patterns in training data. Tools: ChatGPT (text), Midjourney (images), GitHub Copilot (code).

⑪ A training dataset is the collection of examples used to teach AI model. Quality is critical because the model learns whatever is in the data. If the data is biased or wrong, the model can produce unfair or inaccurate results (e.g. discriminating against certain groups).

⑫ Computer vision is the field that lets machines "see" and understand images or videos. They break pictures into pixels and patterns, then recognise objects.

Use cases: self-driving cars detecting obstacles, and facial recognition for unlocking phones

⑬ SECTION C: APIs TOOLS & Developer Concepts

⑭ An API is a set of rules that lets different software systems talk to each other.

Simple analogy: a waiter in a restaurant takes your order (request) to the kitchen (system) and brings back food (response). APIs are important because they let apps share data

and services without needing to know each other's internal details.

②0 An API key is a secret code that identifies and authorises you when using an API. It should be stored securely (environment variables or secret managers), never hard-coded or shared publicly. If exposed, others can use your account, run up costs or access private data.

②1 ~~REST~~ → Representational State Transfer. It is a common style for building APIs.
Main methods:

≡ GET: retrieve data

≡ POST: create new data

≡ PUT: update existing data

≡ DELETE: removes data

②2 JSON is a simple text format for storing and exchanging data. It is easy for both

humans and machines to read. so it is the standard for most APIs.

EXAMPLE { "name": "Thabo",
"age": 28
"city": "Johannesburg" }

② Prompt engineering is the skill of writing clear, specific instructions to get better results from AI tools.

Poor: "Tell me about dogs."

Better: "give me 5 ^{fun} facts about Golden Retrievers."

Poor: "Write an email"

Better: "Write a polite 3-sentence email to a client apologising for one day delivery delay."

④ A map function takes a list and applies the same operation to every item, returning new list.

Python example: numbers = [1, 2, 3]

doubled = list(map(lambda x: x * 2, numbers)) # [2, 4, 6]

25) Cloud computing means using servers and services over the internet instead of your own hardware.

- IaaS (Infrastructure as a Service): environment to build and run apps (Google App Engine)
- SaaS (Software as a Service): ready-to-use applications (Microsoft 365 or Gmail)
- PaaS (Platform as a Service): env virtual machines and storage (Amazon EC2).
- PaaS (Platform as a Service): environment to build and run apps (Google App Engine).

26) A version control system tracks changes to files overtime.

Git is the most popular one because it is free, fast and works well for teams.

A commit is a saved snapshot of changes with a message.

A branch is a separate line of work so you can try ideas without affecting the main code.

②7 A Jupyter notebook is an interactive document that mixes code, text and results. It is most commonly used with Python.

Data Analysts like it because they can write code, see results immediately, and explain their thinking in one place.

②8 Python is a clear, readable programming language that is easy to learn. It has a huge number of libraries for data and AI, so it becomes the standard. Common Libraries are:

- ≡ Pandas: working with tables and data cleaning
- ≡ NumPy: fast numerical calculations
- ≡ Matplotlib/Seaborn: Creating charts
- ≡ Scikit-learn: machine learning models

SECTION D: Data Ethics, Governance & Careers

②9 Data privacy is the rights of people to control how their personal information is collected and used. It is important to control individuals from misuse & harm. GDPR is a

European Union law that sets strict rules for handling personal data. It applies to any organisation that processes data of people in the EU, even if the company is outside EU.

③ Data Ethics is about using data and AI in ways that are fair, transparent, and respectful.
Examples: • Selling personal data without consent
Using AI that discriminates in hiring/lending,
Collecting location data secretly

③ Bias in AI means Data governance is the set of rules, roles and processes that make sure that data is accurate, secure, and used properly.
Key components: clear policies, defined ownership, data quality standards, security controls and regular monitoring

③ Bias in AI means the model systematically favours/disadvantages certain groups because

of the data it was trained on.

Real world example: facial recognition systems that performed worse on dark-skinned faces. Consequences can include unfair arrests or denied opportunities.

③ Roles	Focus	Skills	Tools
Data Analyst	Reports, dashboard insights	SQL, Excel, visualisation	Power BI, Tableau, SQL
Data Scientist	Advanced analysis, prediction	Statistics, ML, coding	Python, R, TensorFlow
Data Engineer	Building data pipelines & systems	ETL, databases, cloud	Spark, Airflow, AWS/GCP
AI Engineer	Building & deploying AI models	Deep learning, MLOps	PyTorch, Kubernetes, APIs

③ Business intelligence is the practice of turning data into useful information for decision-making. Power BI and Tableau let people create interactive dashboards and reports so managers can see trends and act quickly instead of guessing.

SECTION E: Personal Reflection

③⑤ I found data lakes to be the most interesting. The idea that you can keep everything in its raw form and decide later how to use it ~~its raw form~~ feels flexible and modern compared to the stricter warehouse approach.

③⑥ Data and AI will probably change most jobs, including mine. I expect tools that automatically clean data or generate first drafts of reports will become normal. In everyday life I already use recommendations and voice assistants, and I think that they will keep getting smarter and more personal.

③⑦ The difference between the various AI concepts (Narrow VS General, ML vs Deep Learning) was the hardest. Watching short explainer videos on Youtube and reading simple blog posts from IBM and Google really came in handy.

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